WILP, BITS Pilani

MACHINE LEARNING · COMPANION READER

Bayesian Learning & Parameter Estimation

Bayes Theorem, Prior/Posterior, Maximum Likelihood Estimation (MLE), MAP, and Loss Derivations

Session 10 · Read alongside the lecture slides · Every slide example worked out in full

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

1 Foundations of Bayesian Learning

Bayes' theorem updates initial prior beliefs using observed training data.

$$P(h \mid D) = \frac{P(D \mid h)\, P(h)}{P(D)}. \tag{1}$$

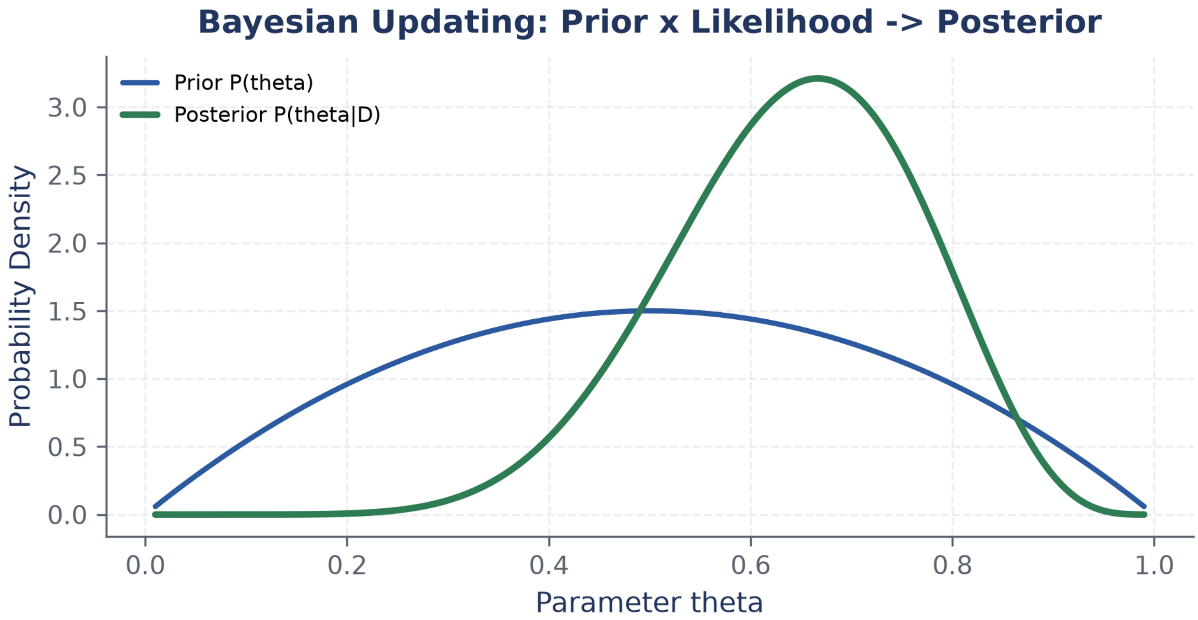

Figure 1: Bayesian updating: Prior distribution updated by Likelihood function.

2 MAP vs Maximum Likelihood Estimation (MLE)

MAP incorporates prior distributions, while MLE maximizes data likelihood alone.

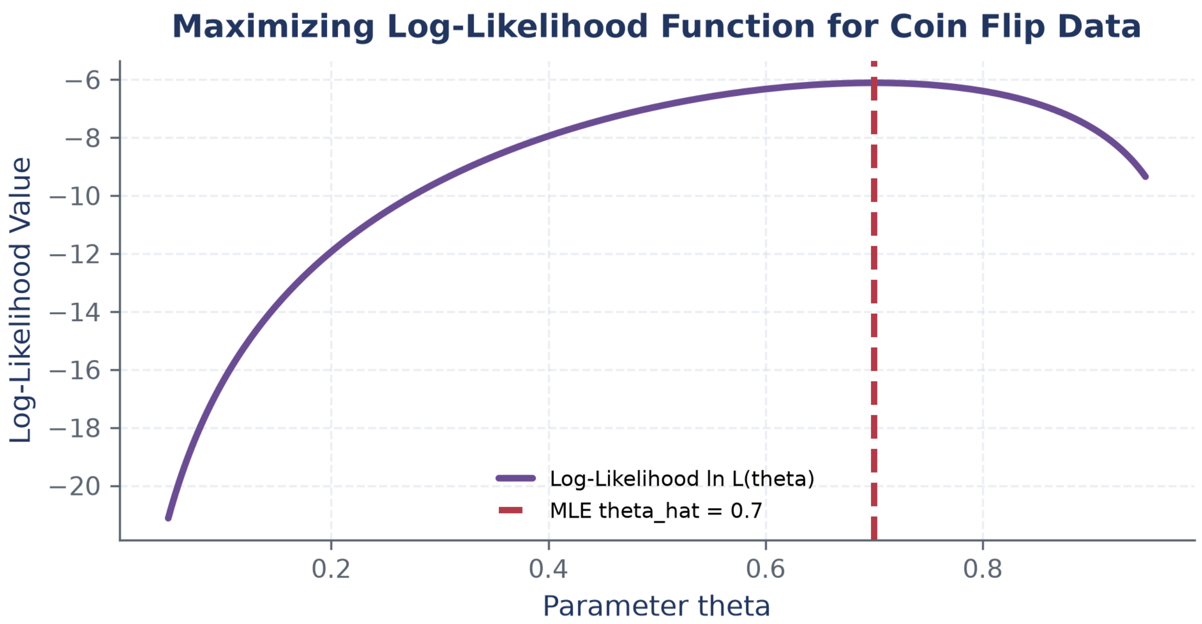

Figure 2: Log-likelihood function optimization.

3 Deriving Loss Functions from Probabilistic Principles

Standard loss functions like Sum-of-Squares are direct consequences of MLE.

Figure 3: Gaussian probability density function.

✎ Worked example: Gaussian Distribution Parameter Estimation via MLE

Given N independent observations drawn from $\mathcal{N}(\mu, \sigma^2)$, derive MLE for mean μ:

Step 1. Partial derivative: $\frac{\partial \ln L}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{N} (x_i - \mu) = 0$.

Step 2. Solve: $\hat{\mu}_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^{N} x_i$. Sample mean is the MLE!